MIL-C-3760F
9 September 1981
SUPERSEDING
MIL-C-3760E
12 December 1973

MILITARY SPECIFICATION

* CLOTH, FLANNEL, WOOL AND COTTON; WOOL AND NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wool and cotton flannel and wool and nylon cloth.

1.2 Classification. The cloth shall be of the following types as specified (see 6.2):

Type I	-	12-ounce, wool and cotton
Type II	-	20-ounce, wool and cotton
Type III	-	12-ounce, wool and nylon

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging and Packing of Woolen, Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

MILITARY

MIL-C-43665 - Cloth Wool: Mothproofing Treatment Of

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF AGRICULTURE

CFR 7 - PART 31 - WOOL STANDARDS

Methods for Determining Grade of Wool Top

(Application for copies should be addressed to the US Department of Agriculture, Agricultural Marketing Service, Washington, DC 20402.)

Rules and Regulations Under the Wool Products Labeling Act of 1939

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

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3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

3.3 Materials (see 6.5).

* 3.3.1 Stock. The blend for yarns shall be not less than 60 percent fleece or pulled wool or a combination of both not lower in grade than 60's US Standard; not more than 10 percent wool noils not lower in grade than 60's US Standard; not more than 10 percent reprocessed, reused, or reworked wool not lower in grade than 60's US Standard; and not more than 20 percent cotton (types I and II) or staple nylon (type III).

3.3.2 Yarn. The yarn for both warp and filling shall be singles, and made from stock which has been blended, carded and spun on the woolen system (see 4.4.3.2).

3.4 Color. The color shall be as specified (see 6.2) and shall be obtained by stock or piece dyeing. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid on the cotton fiber is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4.3.2. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in the test method specified in 4.4.3.2.

3.4.1 Mothproofing. Mothproofing of the material shall be accomplished in accordance with MIL-C-43665 during the stock dyeing or piece dyeing operations, as applicable.

* 3.4.2 Matching. The color of the finished cloth shall match the standard sample under artificial daylight having a color temperature of 7500 $\pm$ 500 kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850 $\pm$ 100 kelvin.

3.4.3 Colorfastness. The finished cloth shall show fastness to wet dry cleaning, crocking, perspiration and light equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished cloth shall show "good" fastness to wet dry cleaning, perspiration and light and shall show a Munsell value for crocking not lower than 8.5. Testing shall be performed as specified in 4.4.3.2.

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* 3.5 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.3.2.

TABLE I. Physical requirements

Type	Width (exclusive of selvage), minimum, inches	Weight (based on linear yard of 56 inch width) minimum, ounces	Yarns per inch minimum		Breaking strength, minimum, pounds		Weave
			Warp	Filling	Warp	Filling	
I	60	12.0	38	32	42	32	Plain
II	60	20.0	46	44	60	50	2/2 Right hand twill
III	60	12.0	38	32	42	32	Plain

3.6 Finish. The cloth shall be scoured, fulled, napped and sheared on the face and on the back. The type and character of finish shall conform to that shown by the standard sample.

3.6.1 pH. The pH value of the water extract of the finished cloth shall be not less than 4.0 nor more than 8.0 when tested as specified in 4.4.3.2.

3.7 Shrinkage. The finished cloth shall shrink no more than 6.0 percent in the warp direction nor more than 4.0 percent in the filling direction when tested as specified in 4.4.3.2.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards for types I and III, and not less than 40 yards for type II. Each length shall be put up in full width rolls as specified in PPP-P-1132.

3.9 Fiber content. The finished cloth shall contain not less than 80 percent wool and the remaining fiber content shall be cotton (types I and II) or staple nylon (type III) when tested as specified in 4.4.3.2.

3.10 Wool content label. The cloth shall be labeled in accordance with the Wool Products Labeling Act of 1939.

3.11 Marking. The back of the cloth shall be marked "BACK" at both ends of each roll with letters not less than 1/2 inch in height (see 4.4.3.1.3). The marking shall be clearly legible in any indelible marking medium used commercially.

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3.12 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for appearance, color, and finish defects and shall be tested for characteristics specified in table II. The presence of any defect or failure of any test shall be cause of rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Wool. Examination for grade (see 3.3.1) shall be performed visually by comparison with the applicable US Standard (see 2.1). The wool components shall be examined prior to blending. In the event of dispute resulting from comparison, the grade shall be determined by a width method (wedge).

4.4.1.1.1 Fleece or pulled wool. Fleece or pulled wool shall be examined in scoured form prior to dyeing, picking or carding. The examination for grade shall be performed on a composite sample of 1 pound of this component for each 10,000 pounds or fraction thereof in the lot. The composite sample shall be selected from not less than 10 percent of the bales or bags in the lot. The lot shall be unacceptable if the sample fails to conform to 60's US Standard or better.

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4.4.1.1.2 Noils. Noils shall be examined prior to dyeing and blending. Sampling and conditions for acceptance shall be as specified in 4.4.1.1.1.

4.4.1.1.3 Reprocessed wool. Reprocessed wool shall be examined for grade when reduced to a fibrous state prior to blending with other component fibers. Method of sampling and sample size and acceptance criteria shall be as specified in 4.4.1.1.1.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to assure that only approved wools are used in the required proportions specified (see 3.3.1). The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

4.4.3 End item inspection.

4.4.3.1 End item examination. The end item shall be examined in accordance with 4.4.3.1.1 through 4.4.3.1.3.

* 4.4.3.1.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section II of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet), shall be scored and assigned demerit points as listed in 4.4.3.1.1.1 except that only those knots and slubs which exceed the limits shown on Sears Fabric Defect Scales (see 6.6), D for slubs and D for knots, shall be scored. Defects on the selvage shall not be scored unless the body of the cloth is affected. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3,200 or less	8
3,201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 10.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 15.0 points. If one roll exceeds 15.0 points per 100 square yards, a second sample shall be examined only for individual roll quality examination. The lot shall be unacceptable if one

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* 4.4.3.2 End item testing. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3-1/2 continuous yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The lot size shall be expressed in units of 1 yard. All test reports shall contain the individual values utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Yarn	3.3.2	Visual 1/
Labile sulfur	3.4	2020
Mothproofing	3.4.1	2/
Colorfastness to:		
Wet drycleaning	3.4.3	5622
Crocking	3.4.3	5651
Perspiration	3.4.3	5680
Light	3.4.3	5660
Weave	3.5	Visual 1/
Weight	3.5	5041
Yarns per inch	3.5	5050
Breaking strength	3.5	5100
pH	3.6.1	2811

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or more rolls in the second sample exceeds 15.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.4.3.1.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Degree and character of finish not equal to standard sample
Width less than specified

4.4.3.1.2 Examination for length.

4.4.3.1.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.4.3.1.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on roll tickets. The rolls examined shall be those selected for the examination of individual rolls.

4.4.3.1.3 Examination for back marking and compliance with the Wool Products Labeling Act. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Back stamping missing from either or both ends
Back stamping on wrong side
Letters less than 1/2 inch in height
Not labeled or ticketed in accordance with the Rules and Regulations Under the Wool Products Labeling Act.

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TABLE II. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method
Shrinkage	3.7	5590
Fiber content:		
Wool	3.9	2100
Cotton	3.9	3/
Nylon	3.9	2530

1/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

2/ As specified in MIL-C-43665.

3/ The cotton content shall be calculated as follows:

$$\text{Cotton content percent} = \frac{R}{S} \times 100$$

R = Weight of residual fibers
S = Weight of dry desized specimen

4.4.4 Packaging inspection. Examination shall be made in accordance with the provisions of PPP-P-1132 to determine that packaging, packing and marking comply with the section 5 requirements.

5. PACKAGING

5.1 Put-up and packaging. Put-up and packaging shall be level A, or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1132.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

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5.3 Marking. In addition to any special marking required in the contract, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

* 6.1 Intended use. The type III cloth is intended for use as lining fabric in Cap, Combat, Woodland, MIL-C-1911.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Types of cloth required (see 1.2).
(c) When a first article is required (see 3.2, 4.3 and 6.4).
(d) Color of cloth required (see 3.4).
(e) Minimum length, if other than specified (see 3.8).
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of 3-1/2 yards of the finished cloth. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.6 Defect scale. Sears Fabric Defect Scales are available from Sears Roebuck and Company, Department 733Q, "Fabric Defect Replica Kit", Sears Tower, Chicago, IL 60684.

6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 11

Preparing activity:

Army - GL

Project No. 8305-0462

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Army - MD
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DOCUMENT IDENTIFIER *(Number)* AND TITLE

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